ARTURS MIHAILOVS

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SUMMARY

Computer Science student with practical experience in **software development**, **cybersecurity**, and programming education. Experienced in building software projects using Python, JavaScript, and embedded systems. Elected **Website Manager** for Trevelyan College JCR and previously **Computer Science Academic Prefect**. Strong communicator with **language skills** in **Russian** (fluent).

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C/C++, HTML/CSS, SQL (SQLite)
Frameworks & Libraries: Node.js, Express, discord.py, NumPy, Pillow, pandas, tkinter, asyncio, BeautifulSoup, cryptography, React, Bootstrap
Tools & Platforms: Git, GitHub, VS Code, npm, LaTeX, Linux (Ubuntu, WSL), VirtualBox, Raspberry Pi
Security Tools: Kali Linux, Wireshark, Nmap, CyberChef, Hashcat, Gobuster, curl, Ghidra, SSH
Hardware: ESP32, Arduino, Flipper Zero (Bluetooth, Wi-Fi, RFID, Sub-GHz)
Concepts: REST APIs, asynchronous programming, low-level assembly-style programming, embedded systems, networking, cryptography, cybersecurity (CTFs), OPSEC

PROJECTS

Photography Portfolio Website

Node.js, Express, JavaScript, HTML/CSS, REST APIs

- Built a full-stack photography portfolio website with a REST API using `Node.js` and `Express`.
- Implemented secure event access, admin authentication, and image upload functionality.
- Developed backend endpoints for creating events, managing photos, and serving image galleries.
- Designed a responsive frontend for clients to browse and access event photos.

Embedded Systems & Cybersecurity Projects

C/C++, ESP32, Arduino, Flipper Zero, Raspberry Pi, Linux

- Developed embedded software for ESP32-based microcontrollers and the Flipper Zero to explore hardware interfaces and wireless communication.
- Built prototypes integrating displays, sensors, and networking functionality.
- Recreated the Mayhem GPIO add-on board for the Flipper Zero, sourcing the boards and radio modules, soldering the hardware, and 3D printing a custom case.

Command-Line Password Generator & OPSEC Tool

Python, Linux (WSL), argparse

- Built a Linux command-line OPSEC tool that generates strong passwords from a 400,000+ word list using Python's secure `secrets` module.
- Added options for word, digit and symbol counts, separators, shuffling, leetspeak, and clipboard copy.

TALKS

Cryptography Talk **June 2024**

Classical Ciphers, RSA, Elliptic Curves, and Post-Quantum Threats

- Presented the history of cryptography, from classical ciphers (ADFGVX, Enigma) to RSA and elliptic curve cryptography.
- Explained how advances in quantum computing threaten today's public-key cryptography.

EDUCATION

MEng Computer Science **Sep 2025 - Present**
Durham University
1:1 achieved during Year 1

- Captured all 20 flags (20/20) in the Networks & Systems cybersecurity coursework CTF.
- Scored 95% on the server-side programming component of the Programming module coursework.
- Wrote and optimised a Little Man Computer (LMC) assembly program to minimise execution cycles, scoring 80% in coursework.

A-levels, Mathematics, Further Mathematics, Computer Science **Sep 2023 - Jun 2025**
King Edward VI Grammar School
A*A*A

- Self-taught the specialised Further Mathematics modules independently from school, achieving an A* in every paper done.